

IRBID, Jordan

WMO: 402550

Lat: 32.550N

Lon: 35.850E

Elev: 2031

StdP: 13.65

Time Zone: 2.00 (JOR)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	35.2	38.1	18.4	14.9	52.4	22.6	18.3	54.4	21.0	47.4	18.9	48.4	3.8	130	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.6	94.6	67.0	91.6	66.9	89.4	66.7	74.0	84.0	72.5	81.8	71.3	80.2	7.0	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.4	125.2	77.7	70.0	119.2	76.1	68.8	114.1	75.0	39.2	84.1	37.7	81.5	36.5	80.9	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.6	16.3	14.2	DB	30.9	101.7	1.6	3.2	29.7	104.0	28.7	105.9	27.8	107.7	26.6	110.1	(4)
(5)			WB	28.8	75.5	2.5	2.4	27.0	77.3	25.5	78.7	24.1	80.1	22.3	81.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.1	48.4	51.0	56.1	62.7	71.2	75.5	78.4	79.0	76.2	70.1	60.0	52.2		
	DBStd	12.13	4.94	5.24	6.56	7.91	6.47	4.74	3.26	3.33	3.81	5.44	5.42	5.33		
	HDD50	188	88	44	14	4	0	0	0	0	0	0	3	35		
	HDD65	1931	516	394	289	137	15	0	0	0	0	16	166	398		
	CDD50	5710	38	71	202	384	657	764	880	900	786	623	302	103		
	CDD65	1977	0	1	13	67	206	314	415	435	336	174	15	1		
	CDH74	16202	0	4	89	634	1914	2745	3514	3665	2483	1076	74	4		
	CDH80	6100	0	0	17	200	743	1072	1380	1494	876	313	5	0		
(14) Wind	WSAvg	5.3	4.5	4.7	5.5	5.4	5.4	7.0	7.8	7.1	5.7	3.6	3.3	3.8		
(15) Precipitation	PrecAvg	19.0	3.9	3.6	3.5	1.0	0.2	0.0	0.0	0.0	0.0	0.6	2.0	3.7		
	PrecMax	26.6	10.6	9.7	8.5	7.6	1.5	0.7	0.0	0.0	0.3	2.0	8.4	9.9		
	PrecMin	10.5	0.7	0.9	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.6		
	PrecStd	4.0	2.2	2.2	1.8	1.3	0.4	0.1	0.0	0.0	0.1	0.6	1.8	2.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.9	73.3	81.3	90.7	97.7	96.5	96.3	98.1	97.0	91.7	80.7	72.6		
		MCWB	51.8	55.8	62.6	63.6	65.2	65.5	69.9	68.6	66.0	65.2	61.0	56.7		
	2%	DB	62.4	67.7	75.7	85.3	91.7	92.9	91.7	92.3	91.4	87.0	76.7	67.1		
		MCWB	50.0	53.7	56.7	61.0	63.3	66.8	68.7	69.2	65.8	64.3	59.1	53.6		
	5%	DB	59.2	63.3	71.7	80.9	87.6	89.2	89.3	89.7	87.6	83.3	73.0	63.2		
		MCWB	49.0	50.9	55.0	59.3	62.5	65.7	67.7	69.2	65.9	63.5	57.5	51.6		
	10%	DB	56.5	59.9	67.7	76.9	83.7	86.1	87.2	87.4	84.7	80.0	69.8	60.2		
		MCWB	48.5	49.5	53.0	57.3	61.7	65.0	67.3	68.4	66.3	62.9	56.3	50.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.1	60.5	66.1	69.0	71.2	75.9	77.2	75.5	74.0	72.2	65.7	59.6		
		MCDB	61.2	69.3	79.7	82.0	86.2	88.4	88.4	85.9	83.3	82.3	76.4	67.9		
	2%	WB	52.5	55.4	61.2	64.7	67.4	71.5	73.9	73.5	71.6	69.2	62.8	56.4		
		MCDB	58.2	63.1	71.2	80.4	82.4	82.2	83.3	83.1	80.6	78.0	71.5	62.6		
	5%	WB	51.0	53.0	57.0	61.1	65.2	68.9	72.0	72.4	70.2	67.4	60.6	54.2		
		MCDB	56.3	60.1	68.3	75.9	80.9	80.6	81.1	81.3	78.4	76.1	68.0	59.9		
	10%	WB	49.6	51.0	54.4	58.8	63.4	67.2	70.6	71.4	69.2	66.0	58.9	52.5		
		MCDB	54.6	57.5	64.7	73.4	78.0	79.5	79.6	79.8	77.2	74.6	65.3	57.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.9	13.7	15.3	18.1	20.0	19.9	18.6	18.6	18.0	16.8	15.6	13.6		
		MCDBR	18.0	20.1	21.8	24.3	24.5	22.6	20.2	20.3	21.1	21.2	20.4	17.9		
		MCWBR	10.0	11.0	11.5	11.6	10.8	10.1	9.0	8.4	8.9	10.1	11.1	9.6		
	5% WB	MCDBR	14.5	16.9	19.5	22.1	21.1	19.8	18.6	18.9	17.9	17.7	16.2	13.8		
		MCWBR	9.6	10.8	12.4	12.0	11.1	10.0	8.5	8.1	8.1	9.7	10.1	8.8		
(40) Clear-Sky Solar Irradiance	taub		0.346	0.373	0.412	0.451	0.450	0.386	0.395	0.401	0.391	0.408	0.365	0.349		
	taud		2.325	2.231	2.104	1.999	2.032	2.255	2.240	2.224	2.255	2.190	2.300	2.330		
	Ebn at Noon		273	277	273	267	268	284	281	278	277	260	264	265		
	Edh at Noon		33	40	48	56	55	44	44	44	41	41	33	31		
(44) All-Sky Solar Radiation	RadAvg		915	1160	1616	1987	2342	2611	2533	2315	1961	1476	1094	859		
	RadStd		65	108	106	118	105	45	27	37	52	64	61	61		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (20 neighbors)		+0.79	N/A	-3.11	N/A	N/A	N/A	N/A	-115	+284	+185	

Nomenclature: See separate page